Sequences, Series & Complex Numbers

Section 1: Arithmetic & Geometric Progressions

1 In an arithmetic progression, the sum of the first n terms is given by $S_n = 3n^2 + 2n$. What is the 10th term a_{10}?

a. 59

b. 62

c. 65

d. 320

2 If the second term of a geometric sequence is 12 and the fifth term is 96, what is the first term?

a. 3

b. 6

c. 8

d. 12

3 Evaluate the sum of the convergent geometric series: $\sum_{n=0}^{\infty} 5(-\frac{1}{3})^n$.

a. $\frac{15}{4}$

b. $\frac{15}{2}$

c. $\frac{5}{4}$

d. 5

Section 2: Binomial Expansions & General Terms

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4 What is the constant term in the expansion of $(x^2 + \frac{1}{x})^9$?

a. 84

b. 36

c. 126

d. 9

5 What is the coefficient of x^4 in the expansion of $(1 - 2x)^6$?

a. 240

b. -240

c. 480

d. -480

6 What is the sum of the coefficients in the expansion of $(5x - 4)^7$?

a. 1

b. 128

c. 2187

d. 0

Section 3: Operations in the Complex Plane

7 What is the complex conjugate of $z = \frac{3 - 4i}{1 + 2i}$?

a. $-1 + 2i$

b. $-1 - 2i$

c. $1 + 2i$

d. $1 - 2i$

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8 What is the modulus of the complex number $z = (3 + 4i)(5 - 12i)$?

a. 65

b. 17

c. 13

d. 25

9 What is the polar form of $z = -3i$?

a. $3(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2})$

b. $3(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

c. $3(\cos\pi + i\sin\pi)$

d. $-3(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

Section 4: De Moivre's Formula & Roots

10 Evaluate $(\cos 30^\circ + i\sin 30^\circ)^6$.

a. -1

b. 1

c. i

d. $-i$

11 Find the real part of $(1 - i\sqrt{3})^5$.

a. 16

b. -16

c. 32

d. -32

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12 How many roots of the polynomial equation $z^4 + 16 = 0$ lie in the first quadrant of the complex plane?

a. 1

b. 2

c. 3

d. 4

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